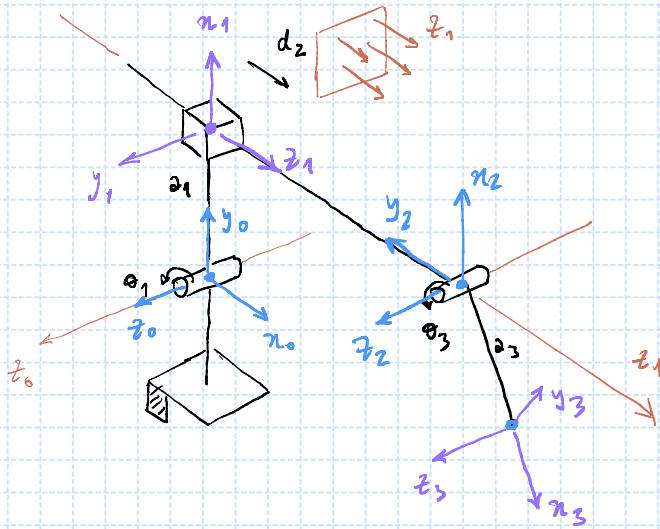


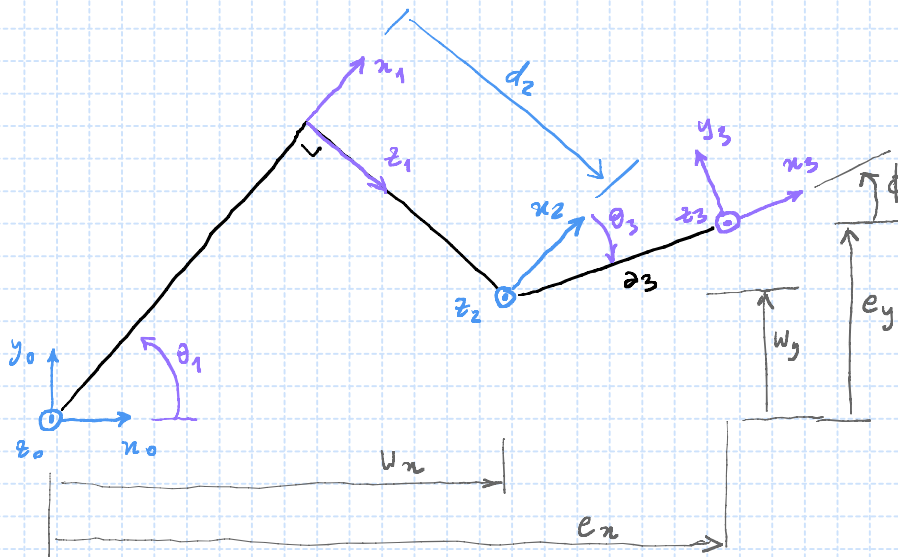
1. (a)



Office Hours

Thursdays: 15h30 - 17h00

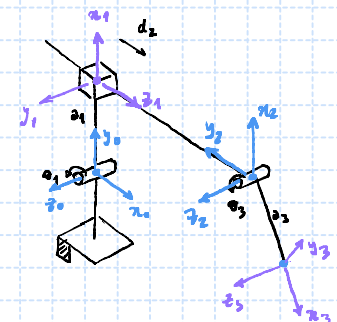
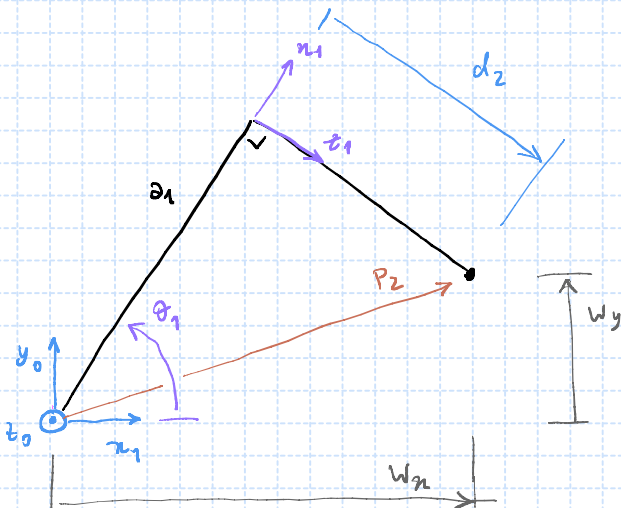
2. (a)

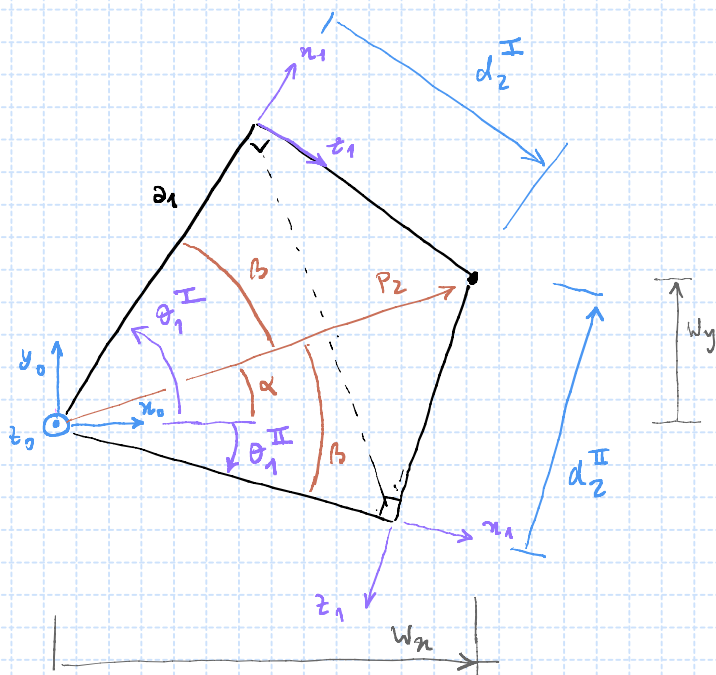


$$(R_2^0)_d = (x_2^0 \ y_2^0 \ z_2^0) = \begin{pmatrix} c\phi & -s\phi & 0 \\ s\phi & c\phi & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$(P_3)_d = \begin{pmatrix} e_x \\ e_y \\ 0 \end{pmatrix}$$

Kinematic Decoupling, $(P_2^0)_d = (P_3)_d - a_3(x_3^0)_d \Leftrightarrow \begin{pmatrix} w_x \\ w_y \\ 0 \end{pmatrix} = \underbrace{\begin{pmatrix} e_x - a_3 c\phi \\ e_y - a_3 s\phi \\ 0 \end{pmatrix}}_{\text{Known}}$





joint 2,

$$\|P_2\| = \sqrt{w_n^2 + w_y^2} = \sqrt{a_1^2 + d_2^2}$$

$$\Rightarrow d_2^2 + a_1^2 = w_n^2 + w_y^2,$$

$$\Rightarrow d_2 = \pm \sqrt{w_n^2 + w_y^2 - a_1^2}.$$

$$\text{solution I, } d_2^I = \sqrt{w_n^2 + w_y^2 - a_1^2}$$

$$\text{solution II, } d_2^{II} = -\sqrt{w_n^2 + w_y^2 - a_1^2} \quad \triangleleft$$

joint 1,

$$\alpha = \arctan(w_y, w_n)$$

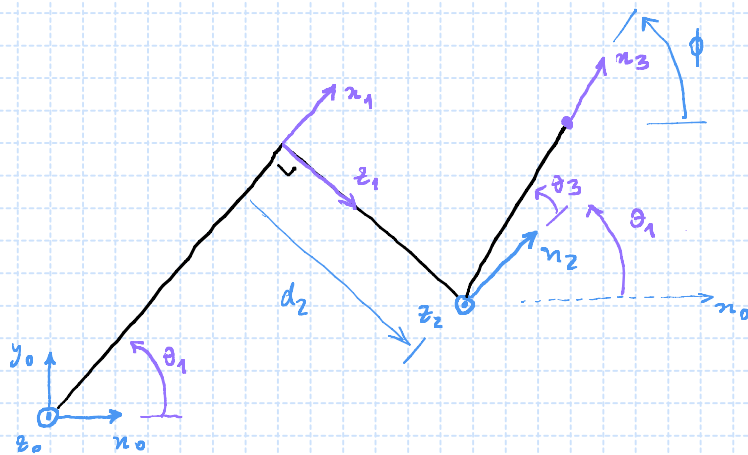
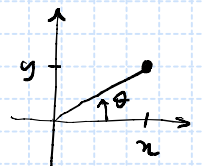
$$\beta = \arctan(d_2^I, a_1)$$

$$\theta_1^I = \alpha + \beta,$$

$$\theta_1^{II} = -(\beta - \alpha) = \alpha - \beta. \quad \triangleleft$$

note 2b:

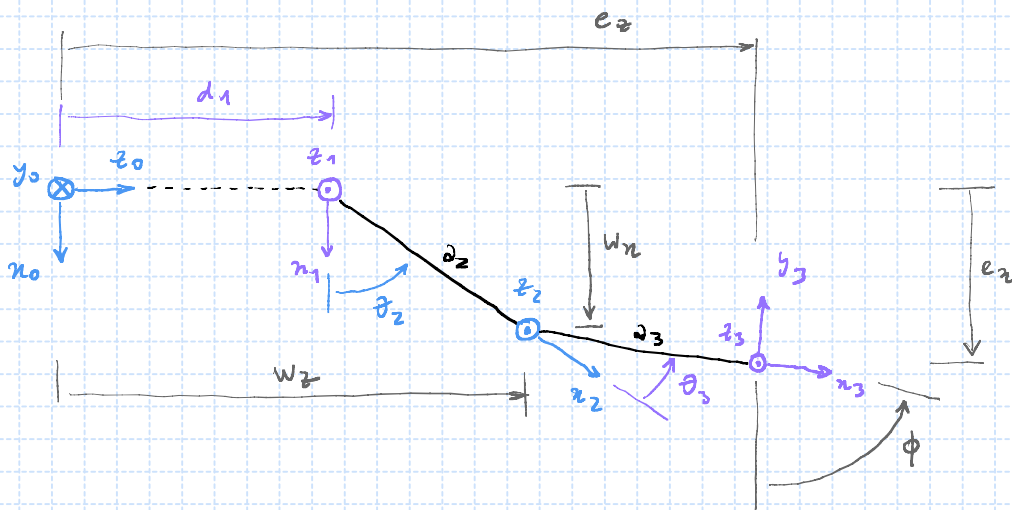
$$\theta = \text{atan2}(y, x)$$



$$\phi = \theta_1 + \theta_3,$$

$$\theta_3^I = \phi - \theta_1^I, \quad \theta_3^{II} = \phi - \theta_1^{II} \quad \triangleleft$$

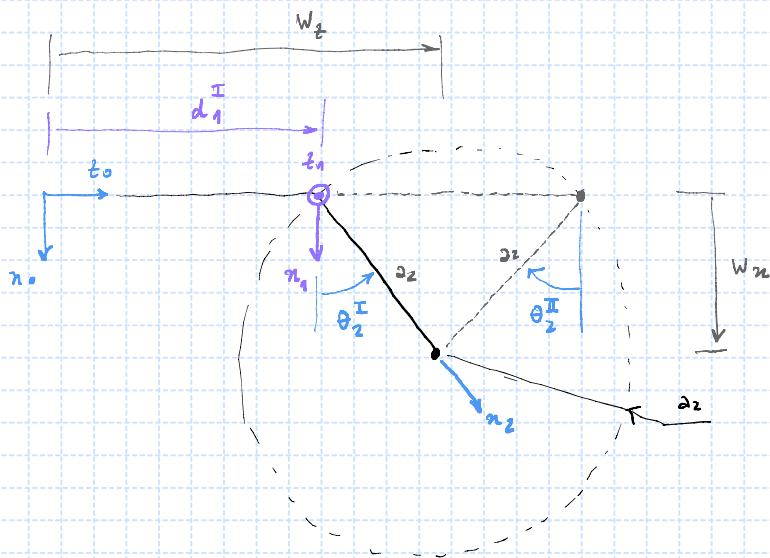
(2) (b)



$$(n_3^o)_d = \begin{pmatrix} c\phi \\ 0 \\ s\phi \end{pmatrix}$$

Kinematic Decoupling,

$$(P_2^o)_d = (P_3^o)_d - a_3 (n_3^o)_d \Rightarrow \begin{pmatrix} w_n \\ 0 \\ w_z \end{pmatrix} = \begin{pmatrix} e_n - a_3 c\phi \\ 0 \\ e_z - a_3 s\phi \end{pmatrix}.$$



$$W_x = a_2 \cos(\theta_2)$$

$$c_2 = \frac{W_x}{a_2}$$

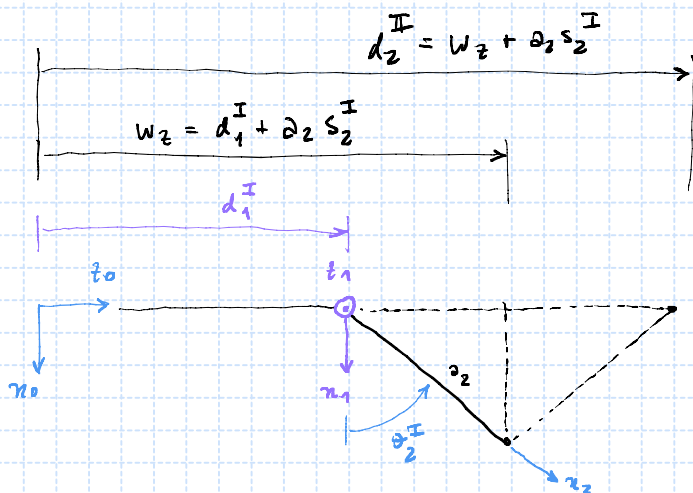
$$\cos^2 \theta + \sin^2 \theta = 1$$

$$s_2^2 = 1 - \left(\frac{W_x}{a_2}\right)^2 \Rightarrow \begin{cases} s_2^I = \sqrt{1 - \left(\frac{W_x}{a_2}\right)^2} \\ s_2^{II} = -\sqrt{1 - \left(\frac{W_x}{a_2}\right)^2} \end{cases}$$

Joint 2

$$\text{Solution I, } \theta_2^I = \arctan\left(\sqrt{1 - \left(\frac{W_x}{a_2}\right)^2}, \frac{W_x}{a_2}\right),$$

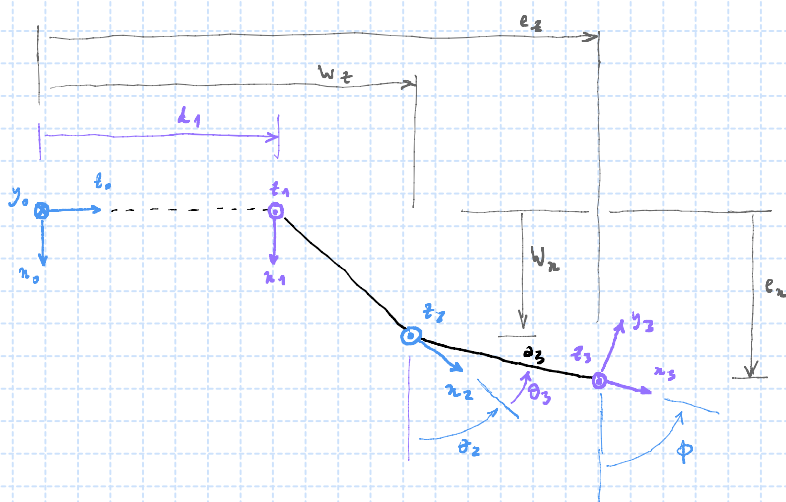
$$\text{Solution II, } \theta_2^{II} = \arctan\left(-\sqrt{1 - \left(\frac{W_x}{a_2}\right)^2}, \frac{W_x}{a_2}\right) = -\theta_2^I. \quad \Delta$$



Joint 1,

$$d_1^I = W_z - a_2 s_2^{II},$$

$$d_2^{II} = W_z + a_2 s_2^{II}. \quad \Delta$$



Joint 3,

$$\phi = \theta_2 + \theta_3$$

Solution I,

$$\theta_3^I = \phi - \theta_2^I,$$

Solution II,

$$\theta_3^{II} = \phi - \theta_2^{II}. \quad \Delta$$